

VIKAS PAL

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Hyderabad, Telangana, India

LinkedIn

GitHub

Portfolio

EDUCATION

Jawaharlal Nehru Technological University Hyderabad (JNTUH) – University College of Engineering

Sultanpur

Sultanpur, Telangana

Bachelor of Technology, Computer Science and Engineering (Lateral Entry) | CGPA: 8.45

2023 – 2026 (Expected)

Quli Qutub Shah Government Polytechnic College

Hyderabad, Telangana

Diploma in Computer Science and Engineering | CGPA: 9.37

2020 – 2023

Gold Medalist – Department of CSE Topper, Batch of 2023

INTERNSHIPS & RESEARCH EXPERIENCE

Research Intern – Machine Learning & Secure Code Analysis (Under Prof. Jyothi Vedurada) IIT Hyderabad, Compilers Lab | Feb 2025 – Present

- Engineered encoding schemes and applied few-shot prompt engineering with DeepSeek to detect insecure code patterns (10 CWE-IDs) using real-world code from web crawlers and the PrimeVul dataset.

Robotics Research Intern (Under Prof. C. Krishna Mohan)

IIT Hyderabad | Sep 2024 – Mar 2025

- Designed and deployed an autonomous UGV navigation system using A* Pathfinding, SLAM, and sensor fusion, achieving 30% faster route planning with reliable real-time obstacle avoidance in GPS-denied terrains.

Open Source Contributor

GSoC (AboutCode), GSSoC (PlayCafe, OnlineBookSales) | Jan 2024 – Nov 2024

- Contributed to AboutCode (GSoC 2023) and multiple GSSoC 2024 projects, improving documentation, code quality, UI, and accessibility — notable issues/PRs: #151, PR #206, #128, PlayCafe #357, OnlineBookSales #408.

Industrial Internship – Java & Spring Boot, Supreme Solution Pvt. Ltd. Hyderabad | Jun 2022 – Nov 2022

- Engineered secure RESTful APIs and microservices with Spring Boot, integrating JWT authentication and CI/CD pipelines using Jenkins, Docker, and Kubernetes.

SKILLS

Programming Languages:	C, C++, Java, Python, JavaScript
Libraries/Frameworks:	React.js, Spring Boot, NumPy, Pandas, TensorFlow, Keras, OpenCV, YOLOv5
Tools/Platforms:	Git, GitHub, VS Code, Docker, Kubernetes, Jenkins, Apache Kafka, Postman, Linux
Databases:	SQL, MongoDB
Software Design & Architecture:	System Design (HLD, LLD)

PROJECTS

AI-Powered Script Generation with Fine-Tuned LLMs | Link

GPT-2, LLaMA, Hugging Face, Gradio

- Engineered and fine-tuned Transformer-based models (GPT-2, LLaMA) on domain-specific data, deploying the solution on Hugging Face Spaces for real-time, scalable script generation.

Smart Emergency Response System | Link

Java, WebSocket, Apache Kafka

- Engineered and developed a real-time emergency alert and tracking system leveraging GPS and mobile technologies to enhance women's safety.

YOLOv8-Based Brain Tumor Detection | Link

Python, YOLO

- Engineered a high-accuracy YOLOv8 model for brain tumor detection in MRI scans using advanced preprocessing and data augmentation techniques.

DevMatch: Developer Networking Platform | Link

Spring Boot, React, MongoDB

- Engineered a full-stack web application with secure authentication, AI-driven skill-project matching, real-time messaging, and advanced search—enhancing developer collaboration success by 60%.

CERTIFICATIONS

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|----------------------------|--------------------------------|-------------------------------------|------------------------------------|
| • Java Foundation – Oracle | • Problem Solving – HackerRank | • AI/ML for Geodata Analysis – ISRO | • Digital Image Processing – NPTEL |
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ACHIEVEMENTS

- Selected for a prestigious one-year Diploma Apprenticeship in Computer Science at DRDO – CHES (Centre for High Energy Systems & Sciences), Ministry of Defence, Hyderabad, under the supervision of Scientist 'F' Mr. Sanjeev Kumar.
- Awarded First Prize for innovative project at the state-level Project Competition at Surjana Tech Fest.
- Technical Coordinator for university-level hackathons and technical workshops.
- Coding Profiles: LeetCode (leetcode.com/u/DDP4LBBXnJ), GeeksforGeeks (geeksforgeeks.org/user/vikas83pal)
- Published a journal article on women's safety solutions titled "Smart Emergency Response System for Women Safety" in IJERT. Available at: [ijert.org/...](http://ijert.org/)